Capturing sunshine to fight climate change: The challenges of photocatalytic CO_2 reduction

Tuning the photoreactor for selective CO production in Luzchem ICH photoreactor

Photocatalytic CO_2 reduction (PCR) has emerged as a potential game-changer in tackling climate change and energy scarcity. PCR offers a clean and sustainable alternative to traditional fuel sources by harnessing sunlight to convert the ever-increasing atmospheric CO_2 into valuable fuels like methane or methanol.

However, despite its promise, PCR faces significant hurdles before reaching widespread commercial application. The main challenge lies in the inherent stability of the CO2 molecule. Breaking the strong bond between carbon and oxygen requires a significant energy input, limiting the overall efficiency of the process.

Furthermore, the success of PCR hinges on the development of highly efficient photocatalysts. These materials play a crucial role in capturing CO_2 molecules, activating them using sunlight, and ultimately converting them into desired fuels. Ideally, the photocatalyst should possess several key attributes

- High CO_2 adsorption: The ability to effectively capture and hold onto CO2 molecules on its surface is essential.
- Strong light absorption: Efficient utilization of sunlight is crucial for driving the photocatalytic reaction.
- Selective product formation: The catalyst should favour the production of specific fuels like methane or methanol, minimizing unwanted byproducts.

Beyond the catalyst itself, the design of the photoreactor, the chamber where the reaction takes place, is also critical. The reactor needs to optimize the interaction between light, CO_2, and the catalyst for maximum CO_2 conversion. Current research efforts are primarily focused on overcoming these limitations by engineering novel photocatalysts and optimising reaction conditions.

Traditionally, research has optimized individual components like photosensitizers, catalysts, and sacrificial electron donors in photocatalytic CO_2 reduction systems. However, it has been recently known that the choice of solvent and additives play significant role in the system's photoredox properties, impacting overall efficiency.

This research introduces a novel Co(II) catalyst designed for CO_2 reduction, emphasizing the use of earth-abundant elements for cost-effectiveness and sustainability. The catalyst is paired with a well-established, earth-abundant photosensitizer to create a fully sustainable system. Furthermore, the study focuses on controlling the selectivity of the Co(II) catalyst. This is crucial for directing the reaction towards desired products, such as valuable carbon-based fuels, rather than solely producing hydrogen gas.

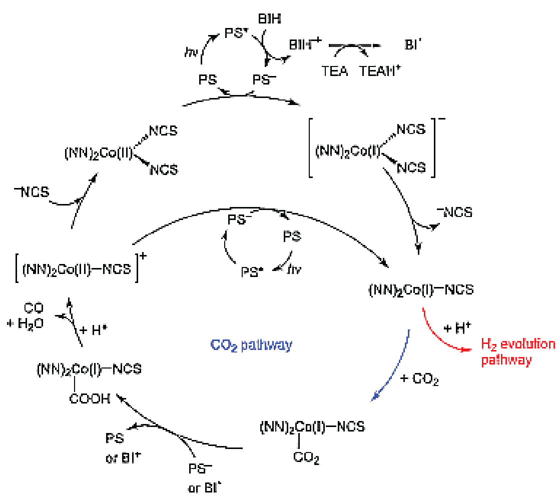

Figure 1: Proposed mechanism for the photoinduced reduction of carbon dioxide with the presented system.

Experimental

The novel cobalt(II) complex 1 was synthesized in dry methanol (MeOH) by mixing in a 2:1 ratio, the chelating diamine ligand, 1-benzyl-4-(quinolin-2-yl)-1H-1,2,3-triazole (BzQuTr) and the cobalt precursor $Co(NCS)_2(py)_4$. The reaction was performed under an argon atmosphere at room temperature. The resulting complex 1 was obtained after evaporation of the solvent, as a lilac precipitate in good yield (60%).

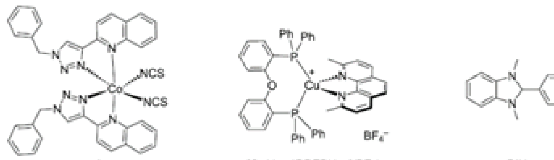

Figure 2: Chemical structures of Co(II) complex 1 as the novel catalyst, the heteroleptic Cu(I) complex as photosensitizer, and the benzimidazolidine derivative BIH as the sacrificial electron donor.

This cobalt complex, designed for CO_2 conversion was structurally characterized using mass spectrometry and elemental analysis. Further confirmations were done using UV-Vis spectroscopy in the chosen solvent (DMA), which revealed its light absorption properties. Cyclic voltammetry explored its electrochemical behavior, suggesting its potential suitability for CO_2 reduction reactions (CO_2RR).

Photo driven CO_2 reduction

The efficiency of the catalyst in photoreduction of CO_2 was analysed. The heteroleptic Cu(I) complex, [Cu(dmp)DPEPhos](BF_4), was selected as a photosensitizer for the development of earthabundant systems. In addition, the benzimidazolidine derivative, BIH (1,3-dimethyl-2-phenylbenzo[d]imidazolidine) was chosen as a sacrificial electron donor, because of its high reducing power. The tests were performed in glass vials (20 mL) equipped with a screw-cap septum. The solutions were prepared under air and CO_2 (or argon) was bubbled inside for at least 10 minutes. TEA or TEOA was distilled twice before use. Experiments were performed in a photoreactor from Luzchem (model: LZC-ICH2) equipped with two lamps at 420 nm (fluorescent lamps of 8 W each) and four mini-stirrers. On each stirrer, two samples were irradiated at the same time, for a total of eight simultaneous reactions. The solutions contained the photosensitizer (1 mM or 0.5 mM), catalyst 1 (different concentrations were studied), and BIH (usually 10 mM or 20 mM). The temperature of the reactor was controlled with an in-built ventilator, T = (25 ± 5)°C. Every test was repeated at least twice. The photon flux was evaluated with actinometry, and it was 0.025 μE s^{-1}.

Therefore, an apparent photoluminescent quantum yield could be estimated to be upto 2.4%, after 4h, according to the equation-

$$\Phi = \frac{\text{CO moles}}{\text{incident photons} \times (1 - 10^{A})} \times 100$$

Where A is the initial absorption value of the photocatalytic system at the irradiation wavelength.

TON and TOF were calculated according to the following equations-

$$\mathrm{TON}_{CO} = \frac{n_{CO}}{n_{catalyst}}$$

$$\mathrm{TON}_{H2} = \frac{n_{H2}}{n_{catalyst}}$$

$$\mathrm{TOF} = \frac{\mathrm{TON}}{t_{reaction}}$$

Where n is the number of moles of the products and of the catalyst and t is the time of the reaction.

HFIP, %	CO, μmol	H_2, μmol	TON_{CO}	TON_{H2}	$Sel._{CO}$
1	4.6	3.1	184	126	59%
2	4.9	2.1	198	83	70%
5	6.0	4.7	231	189	55%

Table 1: Photocatalytic CO2 reduction tests with different concentrations of HFIP- In a 20 mL flask, 5 mL of a solution with the following concentrations, PS (0.5 mM), BIH (20 mM), complex 1 (0.005 mM)and the indicated amounts of HFIP was irradiated for 4h.

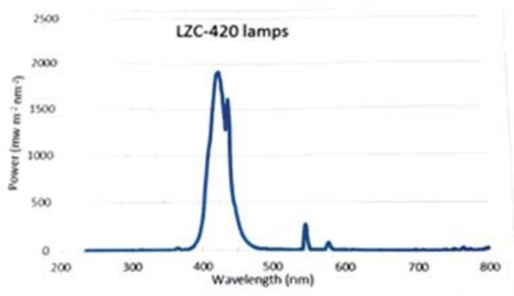

Figure 3: Emission spectrum of the Luzchem lamp at 420 nm.

Conclusion

A fully earth-abundant system was employed for CO_2RR. Initial experiments with a DMA/TEOA solvent mix yielded both CO and hydrogen from CO_2. To favour CO formation within the reactor, the reaction conditions were meticulously adjusted.

This optimization achieved a dominant CO pathway (97% selectivity) through a specific modification to the reaction environment within the photoreactor. Further optimization efforts within the reactor maximized CO production efficiency (TON of 86) after extended light exposure to 15 hours. While introducing a supplemental proton source (HFIP), later enhanced CO evolution (TON up to 230) within 4 hours, it also decreased selectivity within the photoreactor.

References

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ICH Photoreactors